Keep the grid stable

Lower losses. Better damping. Fewer changes.

Participant brief | First draft, version 0.1

1 Your mission

Power grids need to operate efficiently while keeping oscillations under control. Changing a line connection can reduce transmission losses, but can also worsen dynamic stability. Your challenge is to find a good balance with as few structural changes as possible.

In teams of **2–4**, build an **automated Python strategy using VeraGrid** that takes a supplied grid scenario and returns an improved configuration. It should:

1. **Reduce active power losses** through permitted network and dispatch actions.
2. **Maintain or improve small-signal stability** by tuning available controls.
3. **Generalise to unseen scenarios** with different operating conditions and action limits.
4. **Use fewer structural changes** when achieving these objectives.

The challenge is part of TenSyGrid, a research project on tensor methods for analysing converter-dominated grids. Use the supplied simulation and analysis tools to explore candidate solutions and test your strategy.

2 What you may change

Each scenario defines its own permitted actions, parameter ranges and budgets. Depending on the case, these may include line connections or disconnections, model additions or removals, controller tuning, and generator active-power setpoints. Choose only from the options exposed by that scenario; not every action is available in every grid. Keep demand, fixed equipment data and evaluation settings unchanged.

For the structural-change score, each line connection/disconnection or model addition/removal counts as **one equally weighted action**. Replacing a model counts as two: one removal and one addition. The resulting variables and equations are not counted separately. Adjusting existing controls or generation setpoints does not increase this count.

3 Operating requirements

- The AC power flow must converge; bus voltages must remain within $0.95 \leq |V_i| \leq 1.05$ p.u., with no branch above its thermal rating.
- Physical small-signal modes must satisfy $\max_i \Re(\lambda_i) < -10^{-5}$ s^{-1}. The evaluator excludes the neutral reference-angle mode where applicable.
- All oscillatory modes above 0.05 Hz must have damping ratio $\zeta_i = -\Re(\lambda_i)/|\lambda_i| \geq 5\%$.
- Respect the scenario's action and candidate-evaluation budgets. Use local validation to check constraints and any supplied RMS disturbance tests.

4 How to approach the day

Start by running the unchanged baseline and reading its metrics. Then improve losses and damping, checking both after each change. Finally, test across several scenarios and remove actions that do not help. The morning focuses on topology and dispatch; midday on dynamic stability; the afternoon on automation and testing. Mentors are available for questions about the models, tools and rules.

5 How your solution is evaluated

The final evaluation runs the same submitted strategy on **100 hidden scenarios**. All teams receive the same evaluation cases. Each case has its own initial grid, permitted actions and maximum structural-change count n_{max}.

$$\text{Score} = 0.35\,\mathcal{S}_{\text{damping}} + 0.25\,\mathcal{S}_{\text{loss}} + 0.25\,\mathcal{S}_{\text{robustness}} + 0.15\,\mathcal{S}_{\text{changes}}.$$

Criterion	Weight	What earns points
Dynamic stability	35%	Higher minimum oscillatory damping, up to a 10% reference.
Loss reduction	25%	Lower active power losses than the same scenario's baseline.
Robustness	25%	More scenarios meeting stability and operating requirements.
Minimal grid changes	15%	Fewer structural actions relative to the scenario's maximum.

The per-scenario damping, loss and change scores are averaged over the evaluation suite. Robustness is the fraction of scenarios that pass the stability and operating checks. The criteria are defined as follows:

$$s_{\text{damping}} = \min\left(1, \max\left(0, \frac{\zeta_{\text{min}}}{0.10}\right)\right), \qquad s_{\text{loss}} = \max\left(0, 1 - \frac{P_{\text{loss}}}{P_{\text{loss,baseline}}}\right),$$

$$\mathcal{S}_{\text{robustness}} = \frac{N_{\text{pass}}}{N_{\text{total}}}, \qquad s_{\text{changes}} = 1 - \frac{n}{n_{\text{max}}}.$$

An unstable case receives zero damping points. Here ζ_{min} is the minimum damping ratio of the assessed oscillatory modes, and losses are measured by the supplied evaluator. The change count satisfies $0 \le n \le n_{\text{max}}$ by construction. If $n_{\text{max}} = 0$, no structural changes are possible and $s_{\text{changes}} = 1$.

Example: if a scenario allows at most 8 structural changes and you use 2, then $s_{\text{changes}} = 1 - 2/8 = 0.75$. This contributes $0.15 \times 0.75 = 0.1125$ to that scenario's weighted score.

6 Getting started and submission

Use the supplied grids, Python starter interface and local evaluator. First run the baseline, then implement your strategy and compare it across the practice cases. Read each scenario's action catalogue rather than relying on device names or a fixed ordering. Local results help development; hidden cases determine the final ranking.

Submit by the end of code freeze at 16:00:

- Your Python strategy in the supplied submission interface, runnable without manual choices for each scenario.
- A short README with the team name, members, run command and any dependencies.
- A brief explanation of your approach and the practice results you obtained, including its main limitations, for the final presentation.

Submit through the channel announced at kick-off. Keep the evaluator and hidden test assumptions unchanged. The jury may review exceptional technical issues or exploits when determining the final standings. Wrap-up, presentations and awards take place from 16:00 to 17:00.

First draft for review. Final submission details and evaluation conventions will be confirmed before the event.

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